Advanced programming Assignment

ROLL NO: CSB24065

NAME: Gourisankar Bhuyan

Session: Spring **Year:2026**

Question:

Design a library system in Java or Python with:

A base/abstract class LibraryItem (common fields like title, year) An abstract/common method displayInfo()

Create subclasses such as: Book (add author) DVD (add duration/genre)

Your implementation should clearly show:

Use of abstraction (common structure in base class) Method overriding in subclasses Polymorphism using a collection of LibraryItem objects

One additional feature: constructor overloading / default arguments OR class/static counter

Answer:

```
import java.util.ArrayList;

abstract class LibraryItem {
    protected String title;
    protected int year;
    protected static int count = 0;

    public LibraryItem() {
        this("Unknown", 0);
    }

    public LibraryItem(String title, int year) {
        this.title = title;
        this.year = year;
        count++;
```

```java
    }

    public abstract void displayInfo();

    public static int getCount() {
        return count;
    }
}

class Book extends LibraryItem {
    private String author;

    public Book() {
        this("Unknown", 0, "Unknown");
    }

    public Book(String title, int year, String author) {
        super(title, year);
        this.author = author;
    }

    @Override
    public void displayInfo() {
        System.out.println("Book Title: " + title);
        System.out.println("Year: " + year);
        System.out.println("Author: " + author);
        System.out.println();
    }
}

class DVD extends LibraryItem {
```

```
    private int duration;
    private String genre;

    public DVD() {
        this("Unknown", 0, 0, "Unknown");
    }

    public DVD(String title, int year, int duration, String genre) {
        super(title, year);
        this.duration = duration;
        this.genre = genre;
    }

    @Override
    public void displayInfo() {
        System.out.println("DVD Title: " + title);
        System.out.println("Year: " + year);
        System.out.println("Duration: " + duration + " mins");
        System.out.println("Genre: " + genre);
        System.out.println();
    }
}

public class LibrarySystem {
    public static void main(String[] args) {
        ArrayList<LibraryItem> items = new ArrayList<>();

        items.add(new Book("Java Basics", 2020, "James Gosling"));
        items.add(new DVD("Inception", 2010, 148, "Sci-Fi"));
        items.add(new Book("Python Guide", 2022, "Guido van Rossum"));
```

```
        for (LibraryItem item : items) {
            item.displayInfo();
        }

        System.out.println("Total Library Items: " + LibraryItem.getCount());
    }
}
```

Screenshots of the output: